

IPCV assignment idea: Color Clash

Color Clash

A Webcam-Based Augmented Reality Battle Game

1. Game Overview

Color Clash is a real-time, two-player fighting game in which the players themselves become the characters. Using a single webcam, the game transforms the physical environment into a colorful, interactive (if we have the time) battlefield. Instead of controlling separate characters on a screen, players see their own bodies integrated directly into the game through augmented visual effects.

The central concept is to combine the physical movements of the players with a virtual battle taking place directly on the webcam feed. Players use their arms, body poses, and movements to cast spells, dodge incoming attacks, defend themselves, grab consumables or evade environmental hazards. Every action is translated into a visual effect that appears to interact with the other player.

The game draws inspiration from the colorful ink battles of Splatoon, the fast-paced mechanics of 1v1 fighting games, and the visual style of classic arcade games. However, its defining feature is that the webcam feed is the game itself, rather than merely an input device for a separate game environment.

The players are not represented by independent characters that mimic their movements. They are the characters. Their real bodies become the targets of attacks, their movements determine the outcome of the battle in a skill based manner, and the virtual effects are rendered directly onto the live camera image.

The objective is to cover the opponent completely in your own color. Each successful attack changes the opponent's appearance, gradually transforming them into the attacker's color. The first player to completely cover their opponent wins the round.

2. The Core Gameplay Experience

When the game starts, the webcam displays both players standing next to each other, facing the camera. Each player is assigned a distinct color, for example, blue and red.

The live camera feed is enhanced with a layer of digital effects that transforms the players into magical fighters (if we have the time, otherwise simple stick figures with a colour will do fine).

The overlay consists of:

- A colored outline surrounding each player.
- A visual effect attached to each player's face, like a character?
- A color-progress indicator showing how much of the opponent's color has covered them.
- Projectiles and spells that travel through the camera image.
- Impact effects that appear directly on the body of a player when they are hit.
- A dynamic environment that changes color and adds hazards in the midst of battle.

The camera image remains visible throughout the entire game. The players can see themselves and their opponent in real time, allowing them to use their actual surroundings and body positions to understand the game state.

Example of a gameplay moment

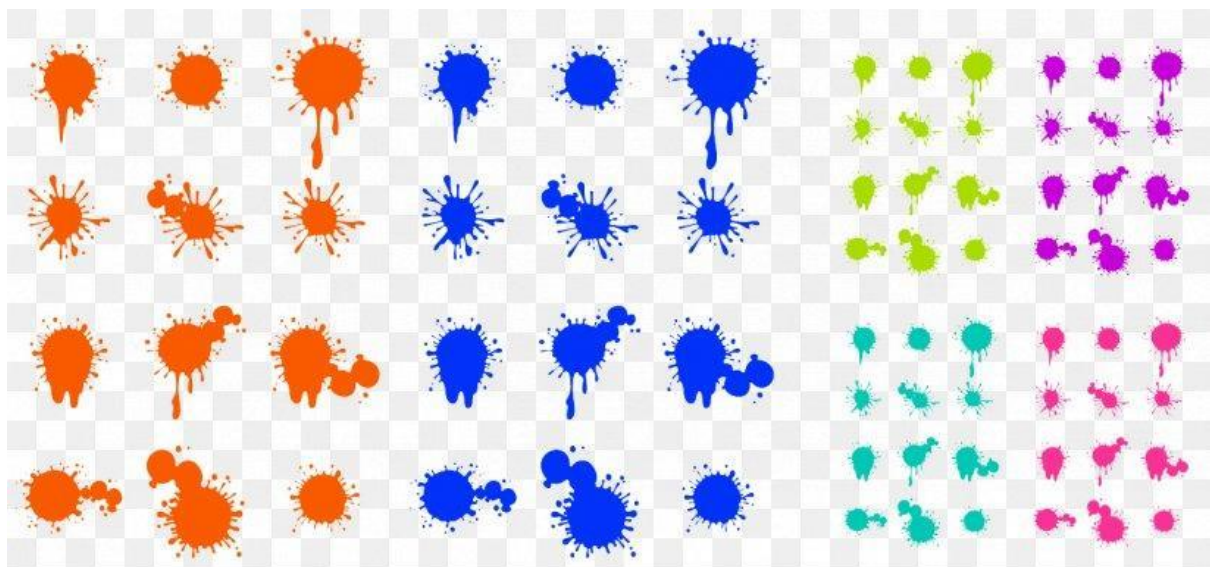
Imagine Player 1, represented by blue, standing on the left, and Player 2, represented by red, standing on the right.

1. Player 1 jabs with their right arm.
2. The system recognizes the movement and generates a blue magical projectile from the player's hand.
3. The projectile travels across the webcam image towards Player 2.
4. Player 2 sees the incoming projectile and ducks to the ground to dodge it.
5. If the projectile misses, it disappears in a small visual effect.
6. If it hits, a blue ink-like splash appears on Player 2's body, and their color progression increases.
7. Player 2 can immediately counterattack by jabbing with own arm.

This creates a continuous interaction in which players must simultaneously observe their opponent, recognize incoming attacks, and coordinate their physical movements to respond.

The result should feel less like playing a conventional video game in front of a camera and more like participating in a magical battle that has been digitally added to the real world.

3. The Visual Design



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The visual style will be deliberately simple and inspired by old-school pixel art. This is both an artistic choice and a practical decision that keeps the project achievable.

Rather than attempting photorealistic effects, we will use bold colors, recognizable shapes, and clear visual feedback.

Player appearance

The players remain visible as real people in the webcam feed. The game adds digital elements to their appearance rather than replacing them with animated sprites.

Each player will have:

- A colored outline or glow that follows their body.
- A face effect that follows their head movements.
- A colored overlay that gradually changes to another color as they take damage.
- Visual feedback when they attack, dodge, or receive a hit.

The colored body overlay is particularly important because it communicates the central game mechanic. As a player takes more damage, their appearance gradually shifts from their original color towards the opponent's color.

This effect could be implemented using a segmented body mask, with the color intensity controlled by the player's damage percentage. A simpler version could use a semi-transparent color overlay within the detected body region.

The battlefield

The background will also be part of the game, but it will not replace the webcam image.

The real environment remains visible, while additional effects create the impression of a magical arena.

For example:

- The area around a player becomes increasingly covered in the opponent's color.
- Successful attacks create colorful splashes on the background.
- Special attacks produce larger effects across the camera view.
- The environment changes to different colours indicating hazards as the battle progresses.

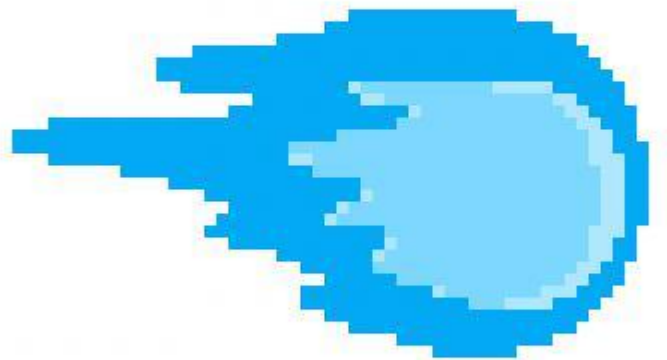
The visual design should ensure that the players and incoming attacks remain easy to distinguish. Visual effects must enhance the gameplay rather than obscure the movements required to play.

4. Combat Mechanics

The combat system is based on a small number of recognizable physical actions. Each action triggers a different attack or defensive ability.

The intention is to make the game easy to understand but challenging to master. Players should be able to recognize incoming attacks and respond through physical movements, rather than relying on random or continuous actions.

4.1 Magic Bolt — Basic Attack



The Magic Bolt is the primary attack used throughout the game.

Activation: jab with one arm.

When a player jabs, a small triangular projectile is generated near their hand and travels towards the opponent.

The projectile is visually associated with the player's color and leaves a short trail behind it.

Gameplay characteristics:

- Fast projectile speed.
- Small hitbox.
- Low damage.
- Short cooldown.
- Can be avoided by moving the body out of its trajectory.

The Magic Bolt is designed to be the most frequently used attack. Its simplicity allows players to focus on timing and dodging rather than learning complicated combinations.

4.2 Energy Beam — Charged Attack



The Energy Beam is a more powerful attack that requires the opponent to react differently from a standard projectile.

Activation: To be discussed how we want to active this ability

After a brief charging animation, a wide beam is generated from the attacking player's position and travels horizontally across the screen.

The beam occupies a larger area and is faster than the Magic Bolt, making it more difficult to avoid. The charging animation however, gives an indication of the timing, making it evadable with skill.

Gameplay characteristics:

- Wider hitbox.
- Higher damage.
- Longer cooldown.
- Brief charging period that provides a warning.
- Much faster.
- Can be avoided by crouching (or jumping)?

The charging period is important because it gives the opponent an opportunity to anticipate the attack. This creates a distinction between fast, low-damage attacks and slower, more powerful ones.

4.3 Meteor Rain — Area Attack



Meteor Rain is an attack designed to control a larger portion of the battlefield.

Activation: Raise both arms simultaneously.

Several small, colored blocks appear at the top of the webcam image and fall towards the players.

The blocks are distributed across a defined region, requiring the opponent to move horizontally to avoid them.

Gameplay characteristics:

- Multiple projectiles.
- Randomized horizontal positions within the target area.
- Low damage per projectile.
- Longer cooldown.
- Can be avoided through vertical movement.

This attack introduces an area-control mechanic, forcing the opponent to consider their position within the webcam image rather than simply reacting to a single projectile.

This attack only has a couple uses per battle. It should also be considered if the other player can attack while someone is trying to dodge these rectangles. If they can attack, the rectangles should be relatively easy to dodge

4.4 Defensive Actions

Dodging is just as important as attacking. Players must be able to use their bodies to avoid incoming spells.

Physical movement Defensive action

Crouch	Avoid horizontal beams
Jump	Invincibility frames
Arms crossing	Activate a temporary shield

A dodge should provide a short period during which the player's hitbox moves away from the incoming attack or becomes temporarily protected.

This allows the game to reward precise timing rather than requiring players to constantly move.

5. The Color-Based Damage System

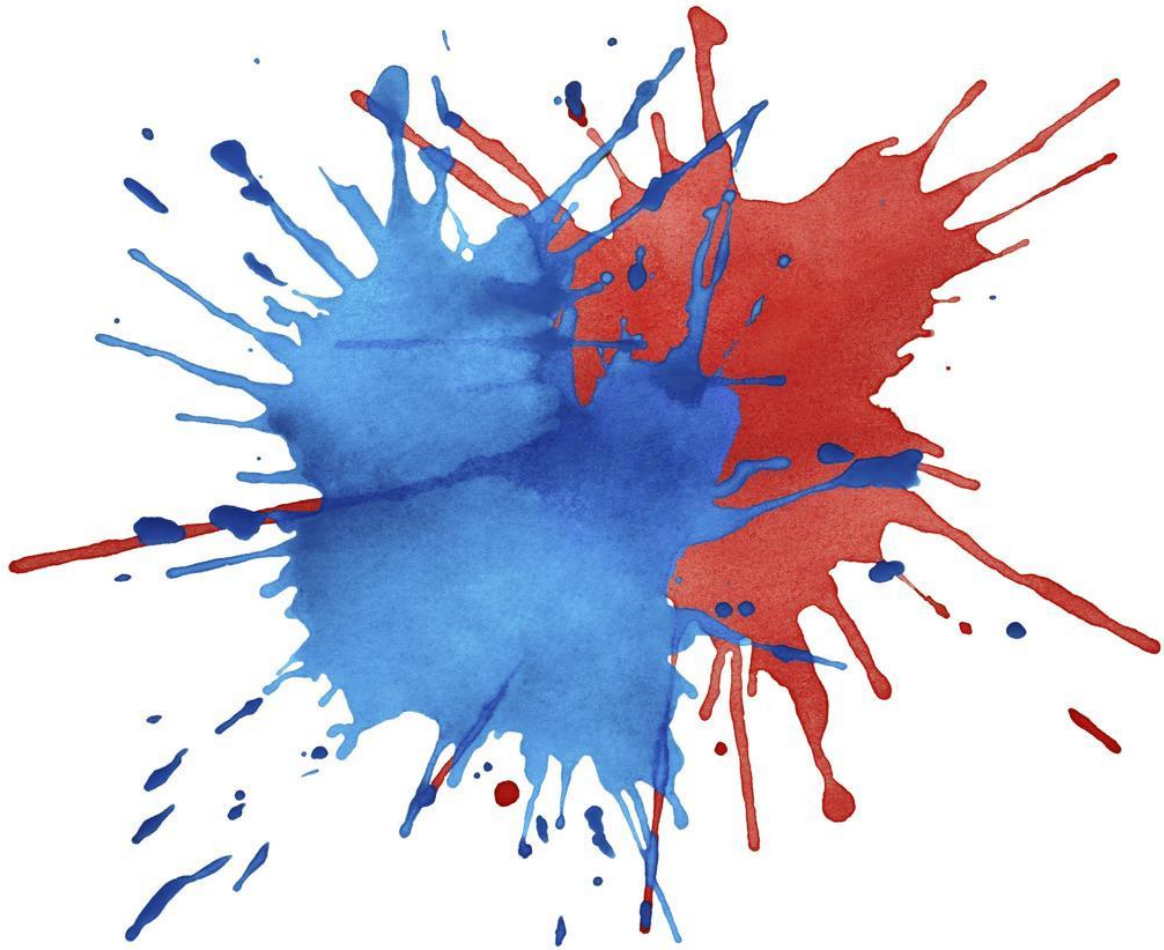
The defining mechanic of Color Clash is its color-based damage system.

Traditional fighting games use health bars to indicate how much damage a player has received. In Color Clash, the damage is represented directly on the player's body.

Each player starts with their own color:

- Player 1: Blue.
- Player 2: Red.

When a player is hit, their appearance gradually changes towards the opponent's color.



For example, after several successful attacks, Player 2 might have a predominantly red appearance with blue patches covering parts of their body.

The color progression is directly linked to the player's remaining health:

Damage received Visual appearance

0%	Original player color
25%	Small patches of opponent's color
50%	Approximately equal color contribution
75%	Mostly opponent's color
100%	Completely covered in opponent's color; round lost

The visual representation could be implemented in several ways, depending on the available time and the performance of the segmentation algorithm.

Goal: A more realistic ink-splatter effect using masks and randomly generated patches that accumulate over the player's body.

The color progression is not merely a visual effect. It is the primary indicator of the game state and the central win condition.

6. Game Flow

A complete game round will consist of four stages.

Stage 1

Player detection and calibration

The webcam activates and detects both players. Each player is assigned a persistent identity and a color. The system establishes their initial positions and checks whether the required body landmarks are visible.

A short countdown gives the players time to prepare.

Stage 2

The battle

The game begins. Players attack, dodge, and defend using physical movements. Projectiles and effects are rendered directly onto the live webcam image, and successful attacks change the opponent's color.

Stage 3

Victory condition

When one player reaches 100% of the opponent's color, the battle ends. The winning player is highlighted, and a victory animation appears over the webcam image.

Stage 4

Restart

The players can restart the game without restarting the application. Their color levels are reset, and a new round begins.

7. How the Game Uses Computer Vision

The computer vision system is not a separate component that simply provides input to the game. It is fundamental to the entire experience.

The game relies on five integrated computer vision tasks, each contributing directly to what the players see and how they interact.

Technical task	Role in the actual gameplay
Face tracking	Adds effects to the players' faces and provides visual feedback
Body pose estimation	Detects attacks, dodges, crouching, and other movements
Multiplayer detection and identity tracking	Ensures that each player's actions, color, and effects remain correctly assigned
Player interaction and game control	Converts physical movements into spells, defensive actions, and game-state changes

Technical task	Role in the actual gameplay
Scene processing and real-time integration	Combines the webcam, tracking, effects, and game logic into one continuous augmented reality experience

The final output is a live video stream with game elements rendered on top of it. The computer vision pipeline processes the camera feed continuously, while the rendering system updates the augmented scene according to the detected movements and game state.

This means the players are simultaneously the input, the characters, and the central visual elements of the game.

8. Minimum Viable Product and Future Development

The game should be developed around a fully playable core experience, with additional features introduced only when the core system is stable.

Minimum Viable Product

Essential

The MVP must demonstrate the complete concept of Color Clash.

- Two players detected simultaneously through one webcam.
- Live webcam feed as the main game display.
- Augmented effects attached to both players.
- Stable player identities.
- One attack triggered by raising an arm.
- One dodge action triggered by body movement.
- Projectiles rendered directly onto the webcam image.
- Collision detection between projectiles and players.
- Color-based damage progression.
- A dynamic background or color overlay that responds to gameplay.
- A complete round with a winner and restart option.
- All five compulsory technical tasks integrated and evaluated.

Stretch Goals

If time permits

Once the MVP is functioning reliably, we can expand the experience with:

- Energy Beam and Meteor Rain attacks.
- Shields and special abilities.
- More advanced ink-splatter effects.

- More detailed body segmentation.
- Animated character effects.
- Power-ups, use of additional weapons and environmental hazards.
- Improved identity tracking during player crossings.
- More sophisticated gesture recognition.
- Additional arenas and game modes.
- Sound effects and a more elaborate victory sequence.

9. The Intended Final Experience

The ultimate goal is to create a game that is immediately understandable to someone watching it, even without an explanation.

Two people stand in front of a webcam. Their bodies are outlined in different colors, and magical effects follow their movements. One player raises an arm and fires a projectile. The other dodges, then counters with an attack of their own. Colorful effects appear on impact, and the players' appearances gradually change as the battle progresses.

The environment reacts to the fight, the attacks are visually recognizable, and the players can clearly see who is winning.

When one player is completely covered in the opponent's color, the game displays the winner directly on the live camera feed.

The defining feature of Color Clash is that the game does not take place on a screen in front of the players. It takes place on the players themselves, with the webcam transforming their real-world movements into a shared, augmented reality battle.